LAB 1 CHALLENGE: Position State-Space Control

Friday shift - Group 11 (Curva Mare)

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1 Control structure and strategy

1.1 Preliminary Analysis

Initially, the design was based on the standard criteria suggested in the handout. However, the resulting response was not entirely satisfactory: the system exhibited a highly restrictive trade-off between speed and oscillations, failing to fully exploit the physical limits of the motor to minimize the rise time. To overcome these limitations, we directly modified the pole placement of the extended system, aiming for a more aggressive yet controlled configuration.

1.2 Controller Structure and Anti-Windup Mechanism

To address the challenge, a state-space control strategy featuring integral action combined with an anti-windup mechanism was adopted. The inclusion of the anti-windup scheme is essential because, in the attempt to drastically reduce the rise time, the controller frequently demands voltages that exceed the physical limits of the actuator (saturation). Without this solution, the integrator would continue to accumulate the tracking error even when the motor is already operating at maximum power (the windup phenomenon). This would result in massive overshoot and prolonged settling times once the system re-enters its linear operating region. The local feedback loop of the anti-windup effectively "brakes" the integrator during saturation, ensuring it remains responsive as soon as the error decreases.

The control system relies on an extended state model $\Sigma_e = (A_e, B_e, C_e, 0)$ that incorporates the integrated error variable $x_I = \int (y - r) d\tau$ to guarantee asymptotic tracking of step references and the rejection of constant disturbances. The anti-windup mechanism enables the use of high gains, meaning the closed-loop system poles can be placed significantly far from the imaginary axis (i.e., with a highly negative real part).

1.3 Pole Placement Logic

The pole placement logic was tuned to achieve a dominant first-order behavior. Rather than opting for a pair of complex conjugate poles with low damping (typical of a second-order approximation)—which, despite being fast, would introduce oscillations and risk violating the $M_p \leq 10\%$ constraint—we chose to make a real pole predominant. This design choice yields an exponential-type response that is inherently free of oscillations.

2 Controller Parameters

The pole placement was performed using the MATLAB `acker` function. The chosen values for the closed-loop system eigenvalues $(A_e - B_e K_e)$ are as follows:

- **Complex conjugate poles:** $\lambda_{1,2} = -80 \pm 17j$
- **Integrator real pole:** $\lambda_3 = -11$

The resulting implementation parameters are:

Parameter	**Value**
State matrix K	[45.07 0.71]
Integrator gain K_I	392.51
Anti-Windup gain K_W	10

Tabella 1: Calculated controller parameters.

3 Experimental Results (Black-Box Model)

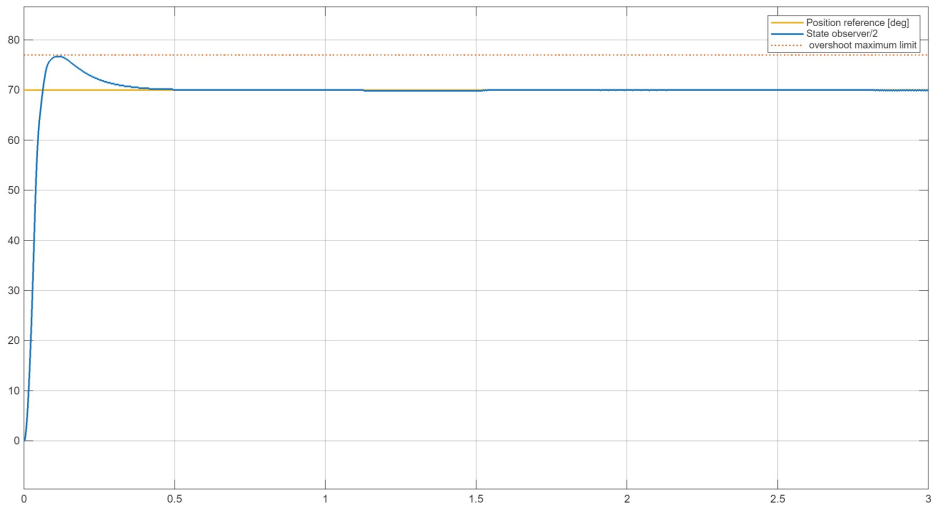

Figura 1: System response to a 70° step input (Black-Box model).

The performance metrics obtained using the `stepinfo` function are summarized below:

- **Rise time ($t_{r,10\%-90\%}$):** 0.037 s
- **Overshoot (M_p):** 9.54 %
- **Peak value:** 76.68 deg

4 Conclusions

The achieved t_r value represents the performance limit reached while ensuring system stability and compliance with the overshoot specifications. The combined use of integral action and anti-windup proved fundamental in maximizing the responsiveness of the Quanser SRV-02 motor.